

Capstone Project - Financial News Sentiment Analysis

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Introduction

need a proper introduction to the topic and THEN relating it to a data science problem

Predicting the movements of the stock market is a long standing challenge. Its erratic and volatile nature leads many to believe that predicting it is a fools game, with even experts in the field unable to consistently outperform the market [6–8]. The Efficient Market Hypothesis (EMH) claims that asset prices quickly reflect all available information [6], it follows that incorporating information and sentiment from sources across the internet may improve the performance of stock price prediction models to address this age old problem. herd behavior and emotions can drive price swings beyond fundamentals [9].

Many efforts have been made to predict stock price movements, with a variety of methods employed, with machine learning being particularly popular in recent years [1].

Sentiment, particularly in recent times on the world wide web present a unique opportunity to leverage in this space. Online financial discourse is heavily concentrated across a few key domains, including social platforms (e.g., Reddit's r/WallStreetBets), news aggregators (e.g., Yahoo Finance), and article-based financial outlets (e.g., Bloomberg), and market announcements [10] . These domains offer continuous textual data that offer differing levels of up to date information and consumer sentiment.

A literature review conducted by [1] analysed 81 papers from January 2015 to June 2020, finding that support vector machines (SVM) were the most popular method, with 26.13% of the analysed papers using them, followed by random forest (RF) (15.32%), and long short term memory (LSTM) (13.51%)

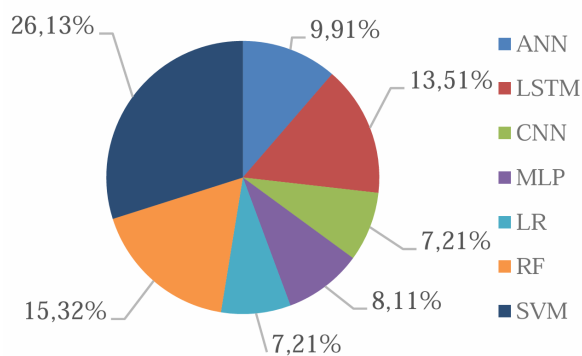


Fig. 1. Proportion of Each Stock Price Prediction method [1]

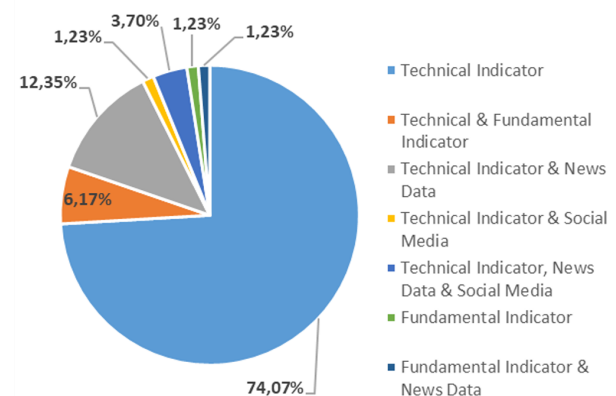


Fig. 2. Proportion of Each of the Studies' Data Source [1]

NLP techniques, not only have been proven applicable to the prediction of stock price, but can also offer value through reducing the burden of manual news monitoring and analysis that is required by a day trader.

This report presents a pipeline that collects news articles using web scraping techniques and processes them through NLP systems to extract meaningful insights. All code and reporting files are available on GitHub.

Web Crawler

Data Type Selection

It is clear that a textual data corpus is required for the exploration of NLP techniques in this setting. There are numerous sources on the world wide web that were considered.

The first source that was considered was the ASX market announcements [10]. The announcements are straight from the source, come at a high frequency, but come with the drawback of sometimes being mundane and not helpful, the format being in a pdf document which is not conducive to text mining with tool such as `BeautifulSoup`, and it being against the terms of use of the ASX [11].

The second source of data that was considered was social media platforms, in particular Twitter and Reddit. Using social media platforms would afford a large quantity of data, but with each document itself being small. Using social media also has the drawback of being unstructured, and also the text data using colloquial language and having spelling mistakes, with extensive preprocessing required as found in many papers [put the papers here, say something about how social media users aren't held to a standard to be relied upon for factual or accurate information](#)

In contrast, financial news articles are more factual than social media, with the added benefit of being longer in length, having correct spelling and grammar. Furthermore, websites can offer more structure in the way they present their articles, with easy ways to scrape key metadata, [perhaps reference the next section where this is elaborated on.](#)

Website Selection

Table 1 shows the comparison of various websites that contain financial and market-related articles, evaluating their suitability for scraping.

Website Name (URL hyperlinked)	Scraping Permitted in Terms of Use?	Allows Article Scraping in Robots.txt?	Uses Cloudflare?	Notes of quality
Market Index	No	Yes	No	Company tags on articles
Yahoo Finance	No	No	No	no mentioned stock labelling. General politics also included which is difficult to relate to individual stocks
Trading Economics	Yes	Yes	No	General politics and economics also included — would require filtering (unstructured)
Financial Review	No	No	No	
The Motley Fool	No	Yes	No	
Sydney Morning Herald	No	No	No	
The Market Online	No	Yes	No	
ABC News	Yes	Yes	No	Regular news mixed with financial news, difficult to use, untagged stock market
Small Caps	Yes	Yes	No	Sitemap highly conducive to scraping. Stock codes provided in articles. All about stocks — no general politics. Static web content (use BeautifulSoup)
Morning Star	No	Yes	No	No sitemap provided
Australian Stock Report	Yes	Yes	No	Unstructured in comparison to other article sites
Stock Head	Yes	Yes	No	Great sitemap but no list of stock codes in article
The Australian	No	No	No	
Hot Copper	No	Yes	No	

Table 1: Comparison of financial and market-related websites for article scraping suitability. Green rows indicate favourable conditions for scraping.

The smallcaps website was selected for the following reasons: - no dynamic content, so no elaborate web scraping web driver is required, can just use `BeautifulSoup` - the website is all about stocks, so no need to filter out non-stock related articles - well labelled articles with stock codes

there is however drawbacks to this particular website and especially only choosing one website from which

to scrape from.

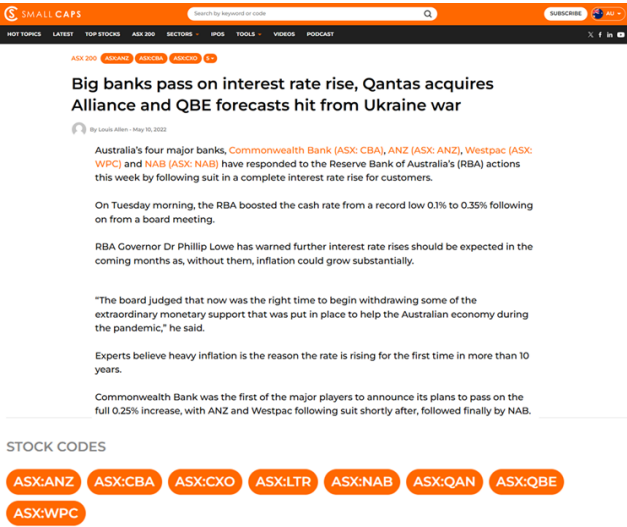


Fig. 3. Example Article [2], containing fields such as title, author, date, and stock codes

asdf

Web Scrapping

To begin this analysis, we scrape the sitemap of `smallcaps.com.au`. This offers the benefit of having a consolidated list of all article URLs, eliminating the need to emulate human navigation through pages of website search results using `Selenium`. The `lxml` parser is used for this task due to its fast performance and capability to handle both XML and HTML scraping. table 2 shows the distribution of the last modified date of the urls in the XML sitemap.

Article Year	Number of Observations
2023	10,468
2024	2,374
2025	323
Total	13,165

Table 2: Number of Observations per Year

Now, individual article content needs to be scraped. The following listing uses `BeautifulSoup` and the `requests` library to extract the contents of a given URL, which was sourced from the sitemap. The `lxml` parser was chosen again for its speed, and since the necessary external C dependency had already been installed with the `lxml-xml` parser during the XML sitemap scraping, no additional setup was required [12]

The contents of the HTML were analysed and parsed into a dictionary capturing the title, author, date, sector, stock codes, and the article body, as demonstrated in the listing below. `BeautifulSoup` was used to extract data from the relevant tags. Below is an example output from the parsing process:

Listing 1: Example HTML extraction code

```
1 soup_test =
  extract_html_from_url('https
  ://smallcaps.com.au/big-banks-
  interest-rate-rise-qantas-
  acquires-alliance-qbe-
  forecasts-hit/')
2 parsed_dict =
  parse_html_to_dict(soup_test)
3 parsed_dict
4
```

Key	Value
'title'	'Big banks pass on int...'
'author'	'Louis Allen'
'date'	'May 10, 2022'
'sector'	'ASX200'
'stock_codes'	['ASX: CBA', 'ASX: ANZ', ...]
'document'	"Australia's four maj..."

Table 3: Parsed dictionary converted into a table

Exploratory Data Analysis

Table 3 depicts the structure of the data. Collected from smallcaps was 1532 articles. Figure 4 depicts the distribution of the scraped articles. The reports sector label is visualised in fig. 6, provided by the smallcaps website. The industry label is provided by the ASX website [3]. It is evident the high proportion of articles in the mining sector, potentially leading to bias in any investigation that is conducted in this data without any other intervention.

Analysing the textual corpus, fig. 5 shows the distribution of document length, where it can be seen that the mean document wordcount is 409 words, which combined with the quantity of data collected should be sufficient for training a machine learning model a source for this would be better than hopes and prayers

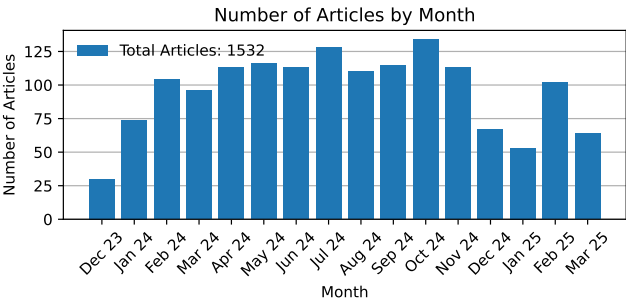


Fig. 4. Quantity of Scraped Articles per Month

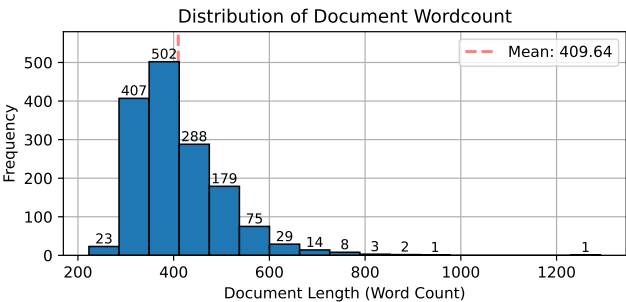


Fig. 5. Distribution of Document Length

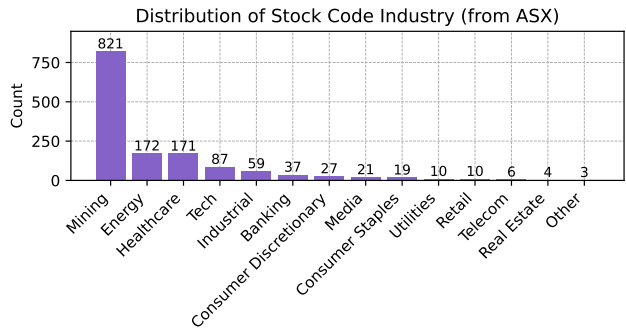
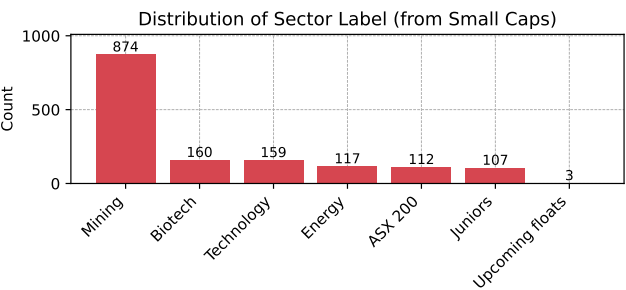


Fig. 6. Distribution of Articles by Sector, using the smallcaps industry label and the ASX sector label [3]

Feature Extraction

Sentiment Analysis

We model the sentiment of an article using the VADER (Valence Aware Dictionary and sEntiment Reasoner), as this has shown to having promising results in a similar setting, as found by [13]. This implementation of vader is assisted through a Geeks for Geeks instructional article [14]

VADER's rule-based approach relies on fixed lexicons and doesn't fully capture the context-dependent sentiment embedded in text, whereas BERT's deep contextual embeddings allow it to discern and represent nuanced sentiment more effectively. FinBERT is a domain-specific adaptation of the BERT model, fine-tuned on financial texts to accurately determine the sentiment expressed in this particular context [15]. This model was retrieved using the huggingface library [16]

While this sounds better, it relies on the data being labelled, making training on a large corpus of data unfeasible. We therefore utilise the finbert-tone pretrained transformer which is the FinBERT model trained on 10,000 manually annotated documents (positive, negative, neutral) sentences from analyst reports.

Machine Learning

After having scraped the articles for the documents **and sentiments?** a machine learning model was developed to predict stock price. figs. 7 and 8 show the structure of a typical feed forward neural network and a recurrent neural network. Recurrent neural networks (RNNs), and others like RNNs, are particularly well suited to time series data

[13] show that an LSTM model, when trained on timeseries price data and sentiment data from news articles, can outperform a model trained on price data alone. Analysis of the literature suggests that SVM models, whilst the most popular [1] often underperformed in a financial setting, due to parameter optimisation being required [1, 17]

Talk about the adam optimiser **According to Kingma et al., 2014, the method is "computationally efficient, has little memory requirement, invariant to diagonal rescaling of gradients, and is well suited for problems that are large in terms of data/parameters". [18, 19]**

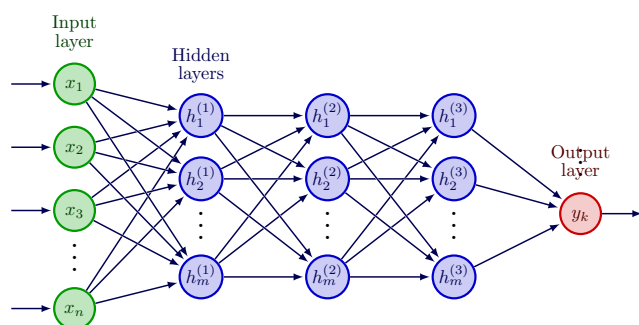


Fig. 7. Typical "FeedForward" Neural Network Topology [4]

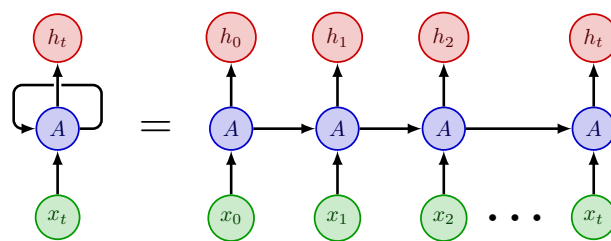


Fig. 8. Recurrent Neural Network Topology [5]

the events extracted from the news are usually quite sparse, resulting in unreliable and unsustainable predictive power [17] (bar for bar) therefore we pivot and try and predict on the asx index which should address sparsity issues, rather than trying to predict volatile single stocks

References

- [1] A. Y. Al-Dulaimi, S. N. Al-Saadi, and S. A. Jasim, “The stock exchange prediction using machine learning techniques: A comprehensive and systematic literature review,” *Jurnal Ilmu Komputer dan Informasi*, vol. 14, no. 2, pp. 132–150, July 2021, accessed: 2025-04-07. [Online]. Available: <https://www.researchgate.net/publication/353002279>
- [2] L. Allen. (2022, May) Big banks pass on interest rate rise, qantas acquires alliance and qbe forecasts hit from ukraine war. <https://smallcaps.com.au/big-banks-interest-rate-rise-qantas-acquires-alliance-qbe-forecasts-hit/>. Accessed: 2025-04-02.
- [3] ASX Limited, “Company directory,” 2024, accessed: 2024-04-08. [Online]. Available: <https://www.asx.com.au/markets/trade-our-cash-market/directory>
- [4] IWN, “Drawing neural network with tikz,” <https://tex.stackexchange.com/questions/153957/drawing-neural-network-with-tikz>, October 2021, answer posted on TeX StackExchange, Accessed: 2025-04-07. [Online]. Available: <https://tex.stackexchange.com/questions/153957/drawing-neural-network-with-tikz>
- [5] user121799, “How do i draw a simple recurrent neural network with goodfellow’s style?” <https://tex.stackexchange.com/questions/494139/how-do-i-draw-a-simple-recurrent-neural-network-with-goodfellows-style>, June 2019, answer posted on TeX StackExchange, Accessed: 2025-04-07. [Online]. Available: <https://tex.stackexchange.com/questions/494139/how-do-i-draw-a-simple-recurrent-neural-network-with-goodfellows-style>
- [6] L. Downey. (2024) Efficient market hypothesis (emh): Definition and critique. Accessed: 2025-04-09. [Online]. Available: <https://www.investopedia.com/terms/e/efficientmarkethypothesis.asp>
- [7] L. Swedroe. (2019) The surprising results from s&p’s latest spiva analysis. Accessed: 2025-04-09. [Online]. Available: <https://www.advisorperspectives.com/articles/2019/11/18/the-surprising-results-from-s-ps-latest-spiva-analysis>
- [8] K. Gunnars. (2024) Is it really true that almost no one can beat the market? Accessed: 2025-04-09. [Online]. Available: <https://stockanalysis.com/article/can-you-beat-the-market/>
- [9] T. Greyling and S. Rossouw, “Twitter sentiment and stock market movements: The predictive power of social media,” <https://cepr.org/voxeu/columns/twitter-sentiment-and-stock-market-movements-predictive-power-social-media>, Mar. 2025, voxEU Column, Centre for Economic Policy Research.
- [10] ASX Limited, “Today’s announcements,” <https://www.asx.com.au/markets/trade-our-cash-market/todays-announcements>, 2025, accessed: 2025-04-02.
- [11] —, “Terms of use,” <https://www.asx.com.au/legals/terms-of-use>, 2023, last updated 12 October 2023. Accessed: 2025-04-02.
- [12] L. Richardson, *Beautiful Soup Documentation*, 2004–2025, retrieved April 27, 2025. [Online]. Available: <https://www.crummy.com/software/BeautifulSoup/bs4/doc/#installing-a-parser>
- [13] C.-R. Ko and H.-T. Chang, “Lstm-based sentiment analysis for stock price forecast,” *PeerJ Computer Science*, vol. 7, no. 1, p. e408, March 2021.
- [14] GeeksforGeeks, “Python sentiment analysis using VADER – using python,” December 2024, accessed: 2025-04-08. [Online]. Available: <https://www.geeksforgeeks.org/python-sentiment-analysis-using-vader/>

- [15] D. T. Araci, “Finbert: Financial sentiment analysis with pre-trained language models,” Master’s thesis, University of Amsterdam, 2019, preprint available on arXiv. [Online]. Available: <https://doi.org/10.48550/arXiv.1908.10063>
- [16] ProsusAI, “Finbert [model],” n.d., retrieved March 27, 2025, from Hugging Face. [Online]. Available: <https://huggingface.co/ProsusAI/finbert>
- [17] X. Zhang, Y. Li, S. Wang, B. Fang, and P. S. Yu, “Enhancing stock market prediction with extended coupled hidden markov model over multi-sourced data,” *arXiv preprint arXiv:1809.00306*, September 2018, 19 pages. Submitted on 2 Sep 2018. Accessed: 2025-04-07. [Online]. Available: <https://doi.org/10.48550/arXiv.1809.00306>
- [18] Keras Team, “Adam class api docs,” <https://keras.io/api/optimizers/adam/>, accessed: 2025-04-07.
- [19] D. P. Kingma and J. Ba, “Adam: A method for stochastic optimization,” 2015, published as a conference paper at the 3rd International Conference for Learning Representations, San Diego, 2015. [Online]. Available: <https://doi.org/10.48550/arXiv.1412.6980>